Bashir Agro Phase 12B: Flutter Farmer App Detailed Instructions

Phase 12B Objective

Build the Flutter Farmer Mobile Application that communicates with the Laravel API.

Scope

Included:
- Flutter project
- Farmer login
- Dashboard
- Task management
- Task submission
- Camera upload
- GPS capture
- Offline support
- Notifications
- Profile

Requires completed Phase 12A APIs.

Flutter Architecture

Recommended:

lib/
core/
features/
services/
models/
widgets/

Features:
- authentication
- dashboard
- tasks
- submission
- profile
- notifications

Authentication Flow

Farmer enters credentials.

Flutter sends request.

Laravel returns token.

Token stored securely.

Farmer enters dashboard.

Dashboard

Display:

- Today's tasks
- Pending tasks
- Completed tasks
- Notifications
- Profile information

Task Workflow

Admin assigns task.

Farmer receives task.

Farmer opens details.

Farmer performs work.

Farmer captures evidence.

Farmer submits completion.

Submission Module

Features:

- Notes
- Camera capture
- Image compression
- GPS capture
- Upload to API
- Submission status tracking

GPS Module

Flutter captures:

- Latitude
- Longitude
- Accuracy

Backend verifies using PostGIS geofence.

Offline Support

Use local storage.

Workflow:

No internet
|
Save locally
|
Connection returns
|
Synchronize data

Notifications

Use Firebase Cloud Messaging.

Events:

- New task
- Reminder
- Approval
- Rejection

Testing

Test:

- Login
- API connection
- Task loading
- Camera
- GPS
- Offline synchronization
- Submission flow

Antigravity Instruction

Implement Phase 12B only.

Build Flutter Farmer App.

Connect with Laravel API.

Do not modify Admin Panel.

Implement authentication, tasks, submission, GPS, offline sync and notifications.

Report:
- Flutter modules
- API integrations
- Tests completed.